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Review

A Review of GNSS Spoofing Detection Based on Multi-Source Fusion

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Abstract

Due to the weak signal power and the publicly known structure of civil signals, Global Navigation Satellite Systems (GNSSs) face an increasingly severe threat from spoofing interference. Constrained by inherent limitations such as physical boundaries and the inability of self-validation, single-source detection methods struggle to effectively counter complex spoofing attacks. Consequently, the advantages of multi-source fusion detection have become increasingly prominent. By incorporating heterogeneous sensors such as Inertial Navigation System (INS), vision, and communication signals, multi-source fusion technology leverages independent reference sources for cross-validation, providing an effective approach to overcome the bottlenecks of single-source detection. This paper systematically reviews 67 representative publications in the field of multi-source fusion GNSS spoofing detection from 2021 to 2026. Existing methods are categorized into three main classes: filtering algorithm-based detection, multi-source data consistency-based detection, and Artificial Intelligence (AI)-based detection. The advantages and limitations of each category are systematically compared, and the current research status along with future development trends are summarized.

Keywords: spoofing detection; multi-source fusion; filtering algorithm; multi-source data consistency; Artificial Intelligence

1. Introduction

GNSSs have deeply integrated into the operational framework of modern society, serving as the core reference for transportation, power and energy, communication networks, and military equipment to acquire high-precision Position, Velocity, and Time (PVT) information. However, due to the weak signals caused by high satellite orbit altitudes and the openly known structure of civil signals, GNSSs inherently possess openness and physical vulnerability, making them highly susceptible to various interference. Among them, the threat of spoofing interference is particularly severe. Unlike jamming interference, spoofing broadcasts forged signals that highly resemble the characteristics of authentic satellite signals, covertly inducing the receiver to generate erroneous PVT results. It can seize navigation control without the user's awareness, leading to catastrophic consequences.

Currently, mainstream single-source spoofing detection methods can be divided into four categories. The first is the method based on spatial geometric features, represented by multi-antenna array Direction of Arrival (DOA) estimation, which identifies spoofing signals from a single source by utilizing the dispersion of satellite signal arrival directions [1–4]. The second is the method based on radio frequency (RF)/baseband physical

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characteristics, which achieves detection by extracting physical fingerprints before signal demodulation, including signal strength monitoring (power, Automatic Gain Control (AGC), and carrier-to-noise ratio (C/N_0) anomalies) [5–9] and Signal Quality Monitoring (SQM, distortion detection of correlation peaks based on multi-correlator outputs) [10–13]. The third is the method based on navigation message and code-layer authentication, represented by Galileo Open Service Navigation Message Authentication (OSNMA) and BeiDou Signal Authentication (SCA), which verifies the authenticity of signal sources by injecting cryptographic signatures at the signal architecture level [14–21]. The fourth is the method based on internal redundancy monitoring of measurements, which utilizes the self-check of solution-logic consistency within the receiver, encompassing Receiver Autonomous Integrity Monitoring (RAIM) [22–24] and clock-consistency monitoring based on internal crystal oscillator characteristics [25,26].

Single-source spoofing detection methods have obvious limitations, mainly reflected in two aspects. First is the constraint of physical limits: for detection schemes based on spatial geometric features and RF/baseband physical characteristics, sophisticated and precisely synchronized generative spoofing can simulate the physical attributes of authentic signals with high fidelity. Through power matching, nanosecond-level code-phase alignment, and distributed collaborative array attacks, the detection sensitivity of methods such as DOA, SQM, and power monitoring significantly degrades or even fails. Second is the constraint of self-validation: for schemes based on navigation message/code-layer authentication and internal measurement redundancy monitoring, single-source systems lack external independent reference benchmarks, making them unable to identify logically consistent false observation data and difficult to defend against meaconing attacks. Furthermore, collaborative spoofing of all visible satellites will cause the RAIM algorithm to fail, rendering it unable to distinguish the cause of the time-pulling effect.

In response to the aforementioned dilemmas of single-source spoofing detection, the advantages of multi-source fusion spoofing detection technology have become increasingly prominent. As the most covert and harmful interference form in GNSSs, spoofing interference has diverse classification forms, which can be divided from multiple aspects such as generation mechanisms, complexity, spoofing strategies, and the number of spoofing sources [27,28]. Different types of spoofing signals have their specific application scenarios and technical characteristics. This complex classification system brings significant challenges to spoofing detection, and current spoofing technologies are continuously developing towards more complex and elusive directions. The core advantage of multi-source fusion technology lies in its possession of independent reference sources, which can effectively circumvent the limitations of single-source detection. By introducing non-satellite source information such as INS, Visual Odometry (VO), Light Detection and Ranging (LiDAR), and land-based radio, this technology leverages cross-validation of heterogeneous data to break through the blind spots of single-source detection. This paper categorizes multi-source fusion GNSS spoofing detection technologies into three main classes, based on technical implementation forms and fusion detection methods. The first category relies on the integrated navigation filtering framework to conduct detection, which is a spoofing identification method derived from the navigation calculation system. The second category detaches from the filtering calculation process, relying on direct mutual verification and comparison of multi-source observation information to complete detection. The third category discards traditional manually defined decision rules, employing intelligent algorithms to mine data features for anomaly identification. The technical implementation paths of the three categories are fundamentally distinct, with clear hierarchical divisions, comprehensively covering existing mainstream research achievements. The specific classifi-

cation structure is shown in Figure 1. The numbers therein represent the specific number of studies.

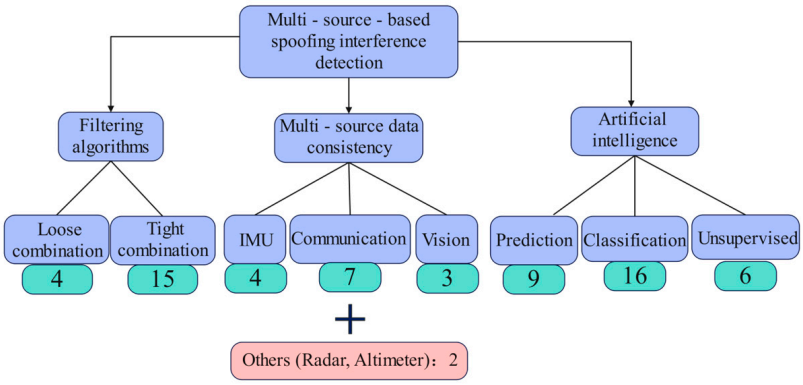


Figure 1. Classification tree diagram of multi-source spoofing detection.

The remaining sections of this paper are arranged as follows: Section 2 presents the literature screening process and reviews existing surveys and previous research achievements related to GNSS spoofing detection; Section 3 focuses on filtering-based detection schemes, detailing spoofing detection technologies from two pathways: loosely coupled and tightly coupled filtering; Section 4 centers on the concept of multi-source data consistency checking, sequentially analyzing corresponding spoofing detection methods from three types of data sources: inertial units, communication signals, and visual information; Section 5 is based on AI technology, collating intelligent detection schemes according to three branches: prediction-based comparison, classification-based recognition, and unsupervised anomaly detection; Section 6 summarizes the current overall research status and prospects the technological development trends; Section 7 concludes the entire paper.

2. Literature Selection and Previous Work

2.1. Literature Selection

To comprehensively grasp the research progress in the field of multi-source fusion GNSS spoofing detection, this study defines the publication timeframe of the selected literature as 2021 to 2026. The search platforms incorporate both mainstream international and domestic academic databases, with the English databases covering IEEE Xplore, Scopus, ScienceDirect, and SpringerLink, and the Chinese databases centering on the China National Knowledge Infrastructure (CNKI), thereby ensuring a comprehensive retrieval of relevant research achievements in this field.

Regarding the search strategy, this study sets nine key search terms for cross-combination, which mainly include spoofing, spoofing interference, navigation interference, navigation interference detection, spoofing interference detection, satellite navigation spoofing interference detection, GNSS interference detection, GNSS spoofing interference, and multi-source fusion spoofing detection. To prevent the omission of relevant literature due to overly singular search terms, explicit synonym expansion rules are formulated. On the one hand, the macro-level term “GNSS” is equivalently expanded to specific system names, such as “GPS” and “BeiDou”, and the generalized term “satellite navigation”. On the other hand, considering the terminology variations and usage habits among different researchers in expressing English technical terms, such as the alternating use of “spoofing” or “deception” for spoofing, and “interference” or “jamming” for interference, this study comprehensively supplements the Chinese and English terminology variations for attack behaviors. Meanwhile, “multi-source fusion” is synonymously expanded to specific technical terms such as “integrated navigation” and “multi-sensor”. Based on these rules,

standardized search queries are constructed for both Chinese and English databases. Specifically, the search query for Chinese databases, taking CNKI as an example, is formulated as: (SU = "GNSS" OR SU = "GPS" OR SU = "satellite navigation") AND (SU = "spoofing" OR SU = "deception" OR SU = "interference") AND (SU = "detection" OR SU = "multi-source fusion" OR SU = "integrated navigation"); the search query for English databases, taking IEEE Xplore as an example, is formulated as: ("GNSS" OR "GPS" OR "satellite navigation") AND ("spoofing" OR "deception" OR "interference" OR "jamming") AND ("detection" OR "fusion" OR "multi-source" OR "integrated").

Through the preliminary query using the aforementioned Chinese and English search formulations, a total of 512 related publications are retrieved. Subsequently, a progressive process of duplication removal, preliminary screening, and detailed rescreening is applied to quantitatively filter and identify the target literature. First, in the duplication removal phase, reference management software Zotero 8 is utilized to identify and remove cross-indexed publications among different databases, excluding 126 duplicate papers and leaving 386 papers. Second, in the preliminary screening phase based on title-subject relevance, screening is conducted directly based on paper titles, which quickly filters out 198 papers that deviate from the core theme of GNSS navigation security, such as radar signal spoofing and pure communication network security, thereby retaining 188 papers that explicitly discuss GNSS spoofing or interference detection technologies. Finally, in the detailed rescreening phase, the remaining 188 papers are further examined for their multi-source fusion characteristics. Among them, 38 papers directly specify features such as multi-source, integrated navigation, or heterogeneous sensor fusion in their titles and are thus directly included in the core candidate library. For the remaining 150 papers that cannot be classified solely by their titles, their full texts are retrieved for in-depth content verification. Through a detailed analysis of the main texts, 28 papers that actually implement multi-source fusion detection algorithms are identified, whereas 122 papers utilizing only a single sensor for detection are quantitatively excluded.

By integrating the final results of the rescreening phase, this study ultimately selects a total of 66 valid research papers with high academic value and reference significance. A quantitative statistical analysis is conducted on the publication years and target application platforms of these selected papers, with the specific distribution of these quantities illustrated in Figures 2 and 3.

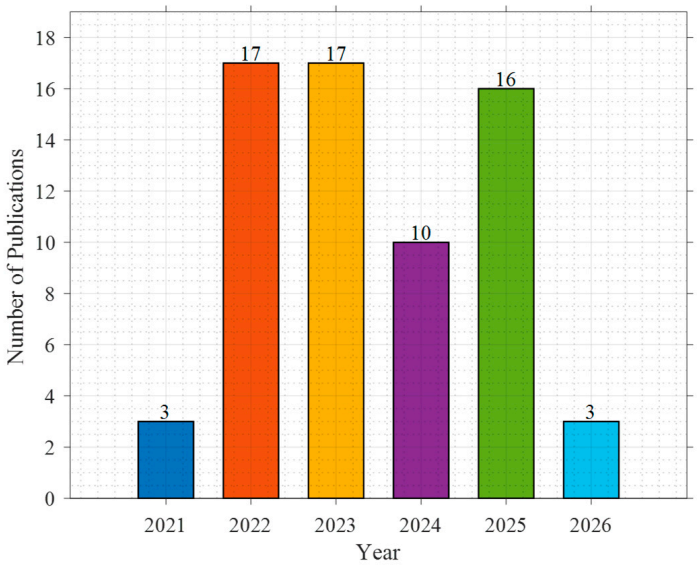


Figure 2. Annual distribution of publications on multi-source fusion GNSS spoofing detection from 2021 to 2026.

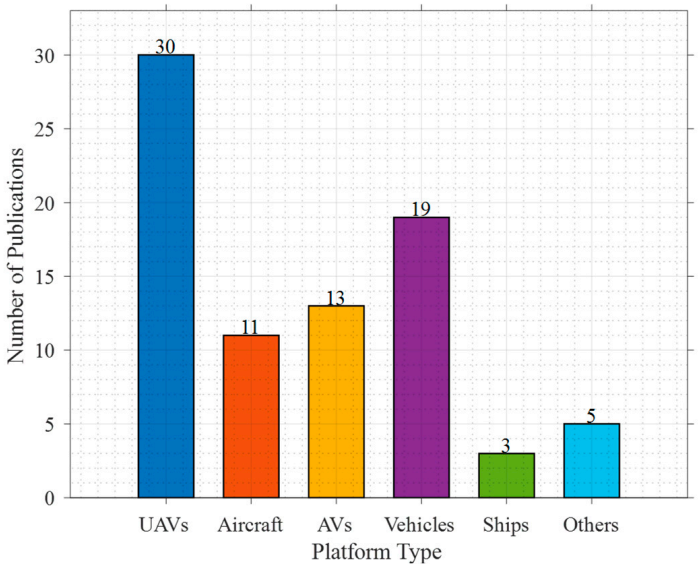


Figure 3. Platform distribution of multi-source fusion GNSS spoofing detection applications.

2.2. Status of Previous Reviews on GNSS Spoofing Detection

In recent years, the literature reviews on Global GNSS spoofing detection technologies have provided systematic overviews of the field. To further clarify the differences among these review papers in terms of their classification logic and research scope, while also evaluating how comprehensively they cover multi-sensor fusion-based detection, this paper selects seven representative review articles. These articles are comparatively analyzed across two dimensions: the taxonomy of detection methods and the comprehensiveness of their coverage of multi-sensor fusion detection. The results of this comparison are summarized in Table 1.

Table 1. Representative review literature on GNSS spoofing detection.

Author	Taxonomy/Classification Criteria	Systematic Review of Multi-Sensor Fusion Detection?
AlAbidy et al. [29]	Classified into two major categories based on technical paths: traditional non-AI detection and AI detection. The AI category is further subdivided into machine learning and deep learning.	No. Only some methods utilize multi-sensor data as inputs, without dedicated classification or systematic review.
Bo Peng [30]	Classified into two major categories based on hardware dependency: ① Hardware-assisted (e.g., antenna arrays, angle/direction of arrival, subspace projection); ② Hardware-free (e.g., Doppler shift, signal consistency, time difference in arrival, statistical parameter analysis, residual signals).	No
Yalan Zhou et al. [31]	Focuses on machine learning-based detection, classified into two major categories based on technical paths: signal classification-based and information verification-based.	Yes. Systematically reviews fusion detection schemes combining machine learning with multi-source information, such as Inertial Navigation System (INS), vision, and cellular signals.

Table 1. *Cont.*

Author	Taxonomy/Classification Criteria	Systematic Review of Multi-Sensor Fusion Detection?
Bertram et al. [32]	Classified into 4 categories based on the machine learning techniques employed: traditional supervised learning, traditional unsupervised learning, neural networks, and other hybrid methods.	No. Only sporadically mentions some studies using multi-sensor or multi-receiver data, without systematically reviewing multi-sensor fusion detection as an independent technical branch.
Shuyan Ni et al. [33]	Classified into 5 major categories based on technical principles: single-antenna signal feature detection, integrated navigation detection, signal spatial correlation detection, machine learning-based detection, and cryptographic authentication detection.	Yes. Dedicated a major category to "integrated navigation-based" detection, systematically reviewing fusion detection schemes of GNSSs with multi-source sensors/navigation methods (such as INS, vision-based navigation, and atomic clocks), while analyzing their performance and applicable scenarios.
Yan Zhou et al. [34]	Classified into 6 categories along the signal processing chain: navigation data information, spatial processing, RF front-end, baseband signal processing, position solutions, and machine learning.	Yes. Under the position solution category, systematically reviews fusion detection schemes assisted by multi-source sensors (such as INS, atomic clocks, and accelerometers).
Radoš et al. [35]	Classified into 5 major categories based on technical principles: signal processing, data bit-level, position measurement, RF fingerprinting, and machine learning/deep learning methods.	No
This Study	Classified into 3 major paradigms: filtering-based spoofing detection, multi-source data consistency check-based spoofing detection, and AI-based spoofing detection. Each category is further elaborated based on technical paths and application scenarios.	Provides broader coverage and more detailed, in-depth technical analysis of multi-source fusion detection methods.

Existing GNSS spoofing detection reviews mostly establish their classification frameworks based on dimensions such as signal processing chains and algorithm types. Consequently, multi-sensor fusion detection is often only sporadically mentioned, typically as a brief, supplementary subsection of traditional detection methods. This lack of systematic classification and in-depth technical analysis makes it difficult to fully present the evolutionary trajectory and implementation details of multi-sensor fusion detection technology. Currently, leveraging the cross-verification of heterogeneous information for spoofing detection has emerged as a major development trend in the field. As research outputs in this area continue to grow rapidly, there is an urgent need for a targeted review to systematically categorize and synthesize these achievements. The core value of multi-sensor fusion spoofing detection essentially stems from the inherent limitations of single-source detection systems and the practical demands of complex application scenarios, which are specifically manifested in the following four aspects:

1. Inherent vulnerability of single sensors: GNSS receivers have limited observable information, and relying solely on their signal characteristics makes it difficult to effectively identify highly self-consistent spoofing signals (such as generative spoofing and meaconing). Particularly when the spoofing signal power is close to the

authentic signal, the sensitivity and robustness of single-source detection methods are severely limited.

2. Challenges of slowly varying and covert spoofing: Single-source detection methods often rely on instantaneous or short-term statistical features, whereas slowly varying spoofing can effectively evade mutation-based detection algorithms by pulling the receiver's positioning results in small steps over a long period. Multi-source detection leverages the short-term high-precision characteristics of autonomous sensors like INS to identify such attacks through long-term accumulated relative deviations.
3. Requirements for system resilience and safety: Modern navigation systems (such as UAVs, autonomous driving, and critical infrastructure) have extremely high requirements for the continuity, integrity, and anti-interference capability of positioning services. Multi-source detection can not only provide an independent positioning reference when GNSSs are completely spoofed but also achieve rapid identification of spoofing and isolation of contaminated observations through multi-source cross-validation, which is a key means to enhance the overall resilience of the system.
4. Limitations in algorithm generalization: Existing single-source detection algorithms are mostly designed for specific spoofing patterns, and their performance drops sharply when spoofing features deviate from design assumptions (such as high synchronization between spoofing and authentic signals, or simultaneous multi-satellite spoofing). Multi-source detection fundamentally reduces the reliance on the inherent features of spoofing signals by introducing physical-level heterogeneous redundancy, possessing stronger generalization capabilities and scenario adaptability.

Therefore, writing a review paper specifically dedicated to multi-source spoofing detection holds significant academic value and practical significance. Compared with existing review papers, this study is grounded in building a more resilient GNSS security defense system, systematically reviewing the current research status and representative methods of multi-source spoofing detection. Its core innovations and primary contributions can be clearly categorized into the following three aspects:

1. Focused and Timely Frontier Scope: By strictly defining the critical time window of 2021 to 2026, which represents a period marked by the rapid advancement of multi-source fusion detection technologies, this study targets, selects, and analyzes the latest core literature, systematically presenting the field's cutting-edge breakthroughs and recent evolutionary trends.
2. Novel Tri-Categorical Classification Framework: Transcending the classification limitations of traditional signal processing chains, this paper innovatively constructs a comprehensive three-category classification framework for multi-source fusion detection. It systematically organizes the vast body of research into three primary technical paradigms: filtering-based, multi-source data consistency check-based, and artificial intelligence-based, thereby offering a clear and structured deconstruction of the multi-sensor detection technology stack.
3. In-Depth Exploration of the Artificial Intelligence Branch: Closely aligned with current research hotspots, this paper systematically categorizes and deeply analyzes three core sub-branches of AI-based multi-source fusion detection. This effectively addresses the theoretical gaps in existing reviews, which often suffer from incomplete classification and shallow analysis regarding the integration of AI with multi-source heterogeneous data.

3. Spoofing Detection Based on Filtering Algorithms

Multi-source spoofing detection methods based on filtering algorithms can be categorized into three types according to the interaction depth and information utilization level

among multi-source data: loosely coupled, tightly coupled, and deeply coupled. In the navigation field, loosely, tightly, and deeply coupled integrations mostly refer specifically to the GNSS/INS integration configuration in industry practice and existing research. Currently, related works implementing spoofing detection relying on filtering also primarily revolve around the GNSS/INS integrated system, with only a limited number of studies adopting GNSSs combined with other sensors to construct the integration framework. Looking at recent research trends, the loosely coupled architecture, which relies solely on GNSS solution results and cannot directly utilize raw observations, has limited detection robustness in complex spoofing scenarios, and related research has gradually become saturated. Spoofing detection methods based on the tightly coupled architecture, by virtue of their ability to directly utilize raw observations such as pseudorange and carrier phase, can perceive signal-level anomalies earlier and have become the current mainstream research direction. In contrast, the deeply coupled architecture, involving deep coupling at the receiver baseband signal level, entails high implementation complexity, making engineering explorations even rarer.

Multi-source spoofing detection methods based on filtering algorithms can be classified into loosely coupled and tightly coupled architectures, depending on the depth of interaction and the level of information utilization among multi-source data. In the navigation field, loosely and tightly coupled integrations typically refer to GNSS/INS-integrated configurations in industry practice and the existing literature. Currently, relevant work on implementing spoofing detection within a filtering framework also primarily centers around GNSS/INS-integrated systems, with only a few studies employing GNSS integrated with other sensors to build detection frameworks.

From the perspective of technical characteristics and research evolution, the loosely coupled architecture relies solely on the estimated GNSS PVT (Position, Velocity, and Time) solutions to participate in filtering without requiring access to raw measurements. Although this lowers the implementation threshold, it also prevents the direct capture of subtle anomalies at the signal level. Consequently, its detection robustness is limited in complex and stealthy spoofing scenarios, and related research has gradually saturated. In contrast, the tightly coupled architecture can directly fuse raw measurements such as pseudoranges and carrier phases. This capability allows it to perceive anomalous features in the signal dimension much earlier, providing stronger detection sensitivity and anti-interference capability, making it the mainstream research direction for filtering-based multi-source spoofing detection today.

It is worth noting that in both loosely and tightly coupled architectures, most existing literature constructs detection algorithms using the filtering innovation sequence as the core test statistic. This is especially crucial when dealing with progressive spoofing, which is an advanced attack method where the spoofer precisely synchronizes with authentic signals, takes over tracking loops, and slowly induces parameter deviations at a rate below conventional detection thresholds, keeping the bias hidden in system noise over the long term. For such slow and stealthy anomalies, single-epoch instantaneous detection often struggles to notice them. To address this challenge, mainstream research widely adopts multi-epoch innovation tests. By continuously accumulating and amplifying tiny state deviations over time, this approach effectively counters the stealthiness of ramp-type spoofing, serving as a highly targeted and widely used detection method.

3.1. Spoofing Detection Based on Loosely Coupled Integration

In the loosely coupled architecture, the GNSS receiver independently completes the positioning solution and outputs position/velocity solutions, which are then fused in the navigation filter (e.g., Extended Kalman Filter (EKF)) with the position/velocity derived

from external auxiliary navigation systems (INS being the most typical, which can also be paired with odometers, celestial navigation, etc.). There is no interaction at the raw observation level between the two. Taking loosely coupled GNSS/INS integration as an example, in this architecture the GNSS and INS complete their navigation solutions independently. Only the solved position and velocity results are fed into the filter for error correction, with no interaction involving raw observation data. Its structure is shown in Figure 4.

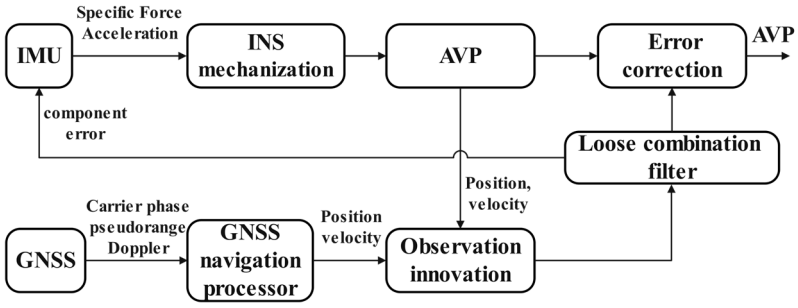


Figure 4. INS/GNSS loosely coupled integration.

In spoofing detection based on the loosely coupled architecture, researchers primarily utilize the anomalous deviations between GNSS position/velocity solutions and inertial propagation solutions for feature extraction and decision-making. Westlund et al. [36], based on the EKF innovation chi-square test algorithm for loosely coupled integration, innovatively introduced a filter recomputation mechanism. This mechanism backtracks to the vehicle's low-dynamic periods to reset the state using INS only, thereby correcting the overly optimistic covariance problem caused by delayed detection and accurately determining the end time of the attack. To address the requirement for fine-grained classification of attack types, Taheri et al. [37] utilized the Interacting Multiple Model Unscented Kalman Filter (IMM-UKF) to model GNSS position manipulations of varying severities as distinct filter modes. By calculating the posterior probability of each mode in real time, they not only achieved spoofing detection but also effectively identified the attack type and estimated its severity. Furthermore, targeting multi-constellation receivers, Akçay et al. [38] employed an accelerometer-assisted loosely coupled filter to cross-compare the normalized residuals of each GNSS constellation against the INS-propagated position. This approach accurately locates and isolates the specific spoofed constellation, thereby maintaining the availability of the navigation solution even when a subset of the system is under attack. To address the degradation in detection sensitivity caused by the long-term drift of inertial propagation, Gu et al. [39] extended the auxiliary measurement source to Visual-Inertial Odometry (VIO). Under a graph optimization framework, they monitored the deviations between VIO position estimates and GNSS position measurements, utilizing an improved chi-square test to trigger alarms. Concurrently, they innovatively introduced a GNSS confidence factor to dynamically adjust the weight of GNSS observations in the integrated filter. This approach not only effectively detected induced spoofing with low injection rates but also ensured that the system maintained high-precision positioning solutions even under attack.

3.2. Spoofing Detection Based on Tightly Coupled Integration

The essential difference between tightly coupled and loosely coupled integrations lies in that the former directly utilizes raw GNSS observations such as pseudorange and pseudorange rate to participate in filtering fusion, rather than relying on the pre-calculated position and velocity information of the receiver. This fusion approach at the raw observa-

tion level retains more sensitive and robust detection information. Taking the GNSS/INS tightly coupled integration as an example, its structure is shown in Figure 5.

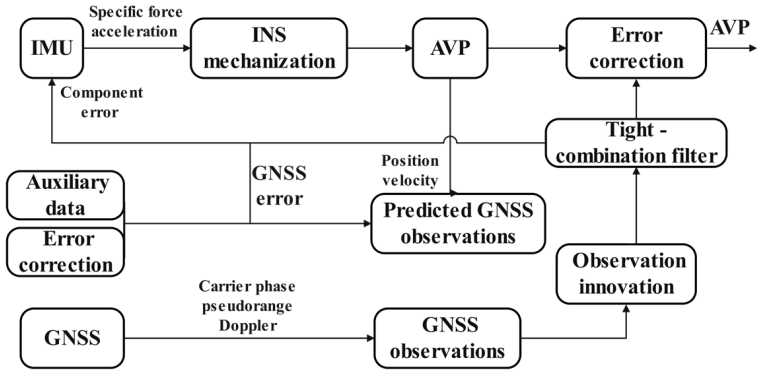


Figure 5. INS/GNSS tightly coupled integration.

Traditional innovation detection algorithms directly use filter innovations to construct test statistics. Due to the error tracking effect of the Kalman filter, they suffer from large detection delays, high false alarm rates, and the inability to promptly determine the end of spoofing when dealing with subtle abrupt spoofing and slowly varying spoofing. To address this, researchers have conducted enhancement studies based on the tightly coupled framework from three dimensions: decision logic optimization in the post-processing stage, data correction in the filter iteration stage, and physical-level heterogeneous information augmentation.

In terms of decision logic optimization in the post-processing stage, researchers are dedicated to improving the sensitivity to subtle deviations by improving the construction method of test statistics, while keeping the internal data flow of the Kalman filter unchanged. Zhong Lunlong et al. [40] introduced the Sequential Probability Ratio Test (SPRT) into the innovation sequence, constructing a log-likelihood ratio statistic to identify spoofed satellites. They also proposed an adaptive SPRT algorithm based on Bayes' theory, which accelerates the accumulation of statistics through adaptive compensation values, significantly shortening the detection time for slowly varying spoofing and improving the detection rate. Zhong Lunlong et al. [41] also introduced the Exponentially Weighted Moving Average (EWMA) control chart into innovation sequence detection, utilizing the weighting of historical statistics and current innovations to construct a test statistic following a Gaussian distribution, which effectively amplifies small deviations in the innovation, shortens detection time, and can promptly detect the end of spoofing. Meanwhile, their method, based on Kullback–Leibler (KL) divergence [42], constructed test statistics by utilizing the potential difference in probability density distributions of the innovation when spoofed versus not spoofed, avoiding the problem of the chi-square test easily expanding the detection region and reducing the missed alarm rate. Fang Xie et al. [43] noticed that the innovation sequence no longer follows a normal distribution under slowly varying spoofing, and proposed a detection algorithm based on a skewness test, which constructs test statistics using innovation skewness, allowing detection utilizing the symmetry breaking of the distribution without requiring error accumulation, reducing the detection delay for slowly varying spoofing by over 35% compared to traditional sequential algorithms. Bai et al. investigated the GNSS/5G tightly coupled system from two dimensions: spoofing detection robustness and post-attack position recovery. Reference [44] focused on solving the false alarms of traditional innovation detection caused by receiver maneuvering; it instead used GNSS single-point positioning results to predict 5G measurements (Time of Arrival/Angle of Departure, TOA/AOD), and performed a chi-square test on the difference

between predicted and measured values as the innovation, achieving maneuver-resistant reliable detection. Reference [45] focused on the position recovery challenge assisted by a single 5G base station; it still adopted the traditional innovation chi-square test to detect attacks, but the core contribution lay in deriving the mathematical model of the EKF output, spoofed position, and true position, and subsequently designing a "coarse plus fine estimation" algorithm to recover the true position with high precision after detection.

In terms of data correction in the filter iteration stage, researchers directly addressed the core defect of innovation contamination caused by the Kalman filter's error tracking, suppressing spoofing impacts and restoring detection sensitivity by intervening in the filtering process or reconstructing the data flow. To prevent the filter from absorbing spoofing signals, Jin et al. [46] proposed a trust factor-based weighted Kalman gain detector, which calculates the weight based on the normalized distance between each satellite's test statistic and the detection threshold. When the statistic approaches the threshold, the gain weight of that satellite's observation is reduced, thereby suppressing the misleading effect of harmful observations on the filtering results and allowing the spoofing deviation in the innovation to accumulate continuously, significantly improving the detection probability of extremely low-drift spoofing. Following a similar logic, Zhong Lunlong, Jiang Yingying, Zhang Chao et al. [41,47,48] introduced a robust estimation mechanism. This method utilizes specific test statistics constructed from innovations (such as EWMA statistics, standard t-distribution statistics, or innovation rates), combined with schemes like IGG-III to design equivalent weight functions, dynamically correcting the Kalman gain matrix. When the test statistic indicates an observation anomaly, the weight of the suspicious observation is adaptively reduced to suppress the contamination and misleading effects of the spoofing signal on the filter state estimation, thereby ensuring the continuous sensitivity of the innovation sequence to spoofing interference. Lyss-Loren et al. [49] proposed a recursive rejection mechanism based on multiple statistical tests, utilizing the chi-square test for overall anomaly perception during the filter iteration stage and combining it with the *T*-test to locate specific spoofed satellites, directly eliminating harmful observations from the filter measurement update. This blocks the contamination of state estimation by spoofing signals at the data source, ensuring meter-level navigation accuracy of the tightly coupled system under multi-satellite spoofing. Furthermore, Zhang et al. [50] proposed a method for reconstructing the prior using multi-epoch INS prediction, introducing an additional INS computation unit that is corrected by the EKF only every N epochs, maintaining an unpolluted prior estimate within the N epochs to construct an improved innovation, drastically shortening the time to alert. Ding Jicheng et al. [51] also constructed a residual sliding variance detection metric targeting spoofing interference with time-delay variation characteristics; instead of directly using the innovation with error tracking, they calculated the variance of the residuals through a sliding window to construct the Residual Sliding Variance (RSV) detection metric, significantly improving the detection response speed.

Regarding physical-level heterogeneous information augmentation, researchers have thoroughly transcended the traditional, singular GNSS/INS internal data framework. Traditional tightly coupled GNSS/INS architectures possess inherent limitations in their anti-spoofing mechanisms; specifically, as a relative measurement device, the INS suffers from cumulative integration drift, meaning the filter must continuously rely on GNSS as an absolute reference to perform closed-loop error compensation on it. However, when encountering slow-growing spoofing, the filter is highly susceptible to misinterpreting the slowly varying deviation induced by spoofing as the physical drift of the INS, consequently utilizing the contaminated GNSS data to falsely compensate the INS and causing the entire system to lose its absolute reference benchmark. To evade this positive feedback contamination trap, the core of the heterogeneous information augmentation mechanism

lies in introducing external references that are free from GNSS closed-loop compensation control and capable of providing independent spatial constraints. For instance, targeting airborne navigation scenarios, Lunlong Zhong et al. [52] proposed a spoofing detection method that fuses multi-source data from INS and distance measuring equipment (DME). Unlike IMUs, DME provides independent absolute ranging based on ground-based beacons, which is free of cumulative integration drift and requires no error compensation from GNSSs. This study leverages the physical independence of DME to construct inconsistency detection metrics, combining them with an improved sequential probability ratio test (SPRT) algorithm based on Bayesian theory to calculate adaptive compensation values, which demonstrates excellent detection response speeds when countering multi-satellite spoofing and minor pseudorange attacks. In terms of vision and environmental perception aiding, Ming-ze Liu et al. [53] proposed a VIO-augmented GNSS spoofing detection scheme. In its system design, this scheme decouples the GNSS correction loop from the VIO, treating the VIO as an independent cross-verification baseline, employing kernel density estimation (KDE) to perform non-parametric modeling on the distribution of positioning differences between VIO and spoof-free GNSSs, and monitoring the deviation of this distribution in real-time using Kullback–Leibler (KL) divergence, which effectively overcomes the strict assumption in traditional detection frameworks that residuals must follow an ideal Gaussian distribution. Targeting the specific application of ground autonomous vehicles, Irfan et al. [54] proposed a sensor fusion detection architecture based on particle filtering. This method further breaks the limitation of relying solely on proprioceptive sensors by deeply integrating vehicle path information with absolute rigid constraints of the physical world, such as road boundaries including lane markings and road edges. By evaluating the likelihood relationship between the observed pseudoranges and the vehicle's probable position distribution estimated via particle filtering within a tightly coupled architecture, the system can effectively identify highly self-consistent pseudorange spoofing attacks forged by adversaries. In summary, by introducing ground-based radio frequency ranging (DME), visual sensors (VIO), and high-definition map road network constraints, the aforementioned studies cut off the path through which spoofing signals contaminate the entire system at the underlying mechanism level, which significantly enriches the research dimensions of heterogeneous multi-source data fusion and effectively addresses the theoretical limitations of previous detection frameworks that relied solely on a single GNSS/INS architecture.

To intuitively present the research progress in spoofing detection based on filtering algorithms, Table 2 categorizes and summarizes representative literature from dimensions such as research platforms and detection methods.

Table 2. Summary of literature on GNSS spoofing detection methods based on filtering algorithms.

Literature	Epoch	Platform	Detection Method
[36]	Multi-epoch	Vehicle	Chi-square test
[37]	Multi-epoch	Naval vessel	Posterior probability detection
[38]	Multi-epoch	Vehicle	Normalized residual multiple comparison
[39]	Multi-epoch	UAV, UV	Chi-square test
[40]	Multi-epoch	Civil aircraft	Sequential probability ratio test
[41]	Multi-epoch	Civil aircraft	EWMA control chart-based innovation weighted statistics + robust estimation
[42]	Multi-epoch	Civil aircraft	KL divergence-based innovation probability density distribution difference detection

Table 2. *Cont.*

Literature	Epoch	Platform	Detection Method
[43]	Multi-epoch	Vehicle	Normality detection based on innovation sequence skewness test
[44]	Single-epoch	Vehicle, UAV	Chi-square test
[45]	Single-epoch	Vehicle, UAV	Chi-square test
[46]	Multi-epoch	/	Chi-square test
[47]	Multi-epoch	UAV	Improved innovation rate direct comparison
[48]	Multi-epoch	Aircraft	Chi-square test
[49]	Single-epoch	Aircraft	Chi-square test + *T*-test
[50]	Multi-epoch	Aircraft, Vehicle	Chi-square test
[51]	Multi-epoch	Aircraft, Vehicle	Residual sliding variance threshold comparison
[52]	Multi-epoch	Civil aircraft	Improved SPRT cumulative statistics threshold comparison
[53]	Multi-epoch	Vehicle	Detection based on KDE and KL divergence
[54]	Single-epoch	Vehicle	Effective particle number threshold comparison

Combined with the above text and Table 2, it can be seen that whether in loosely coupled or tightly coupled systems, spoofing detection methods based on innovations/residuals have shown significant evolutionary trends: In terms of detection mechanisms, to cope with increasingly covert, slowly varying spoofing, single-epoch "snapshot" detection has gradually given way to multi-epoch cumulative "continuous" detection. Among the 19 relevant studies reviewed, only 4 studies explicitly construct detection metrics based on single-epoch information, while the majority adopt continuous methods. In terms of performance optimization, the core focus of research is to counteract the error tracking effect of the filter, striving to significantly shorten detection delays and improve accuracy. In terms of detection dimensions, detection capabilities are expanding from coarse-grained system-level alarms to fine-grained process perception, not only enabling earlier identification of attack occurrences but also accurately determining the end time of attacks, and even achieving channel-level localization and isolation of interfered satellites. This evolution from "single-point to continuous", "existence judgment to full control of start and end", and "system-level to channel-level" marks that the multi-source fusion spoofing detection technology based on filtering algorithms is continuously deepening towards faster, more accurate, and more comprehensive directions.

However, filtering-based methods generally rely on accurate presetting of the prior statistical characteristics of system noise and measurement noise. To avoid the above limitations brought by recursive filtering and integrated navigation, spoofing detection based on multi-source data consistency has received widespread attention. This type of method does not rely on recursive filtering and integrated navigation algorithms; it only utilizes redundant observations of the same motion state from heterogeneous sensors such as GNSS, IMU, communication signals, and vision, and achieves spoofing detection by directly comparing the consistency of multi-source outputs. With advantages such as intuitive principles, strong robustness, and independence from filtering models, it has become an important research branch in the current spoofing detection field.

4. Spoofing Detection Based on Multi-Source Data Consistency Verification

Spoofing detection based on multi-source data consistency verification primarily utilizes the redundancy of multi-source sensors in observing the same physical state to identify spoofing attacks, without relying on filtering algorithms. Since attackers find it difficult to implement full-dimensional collaborative spoofing on heterogeneous sensors, the output data of the spoofed sensor will inevitably deviate from the true physical state, thereby disrupting the consistency among multi-source data. In recent years, detection methods based on direct consistency have been mainly divided into three research branches according to the different auxiliary observation sources they rely on: methods based on IMU, communication signals, and visual information. The following sections will elaborate on these three types of methods, respectively.

4.1. Spoofing Detection Based on IMU

Direct consistency-based spoofing detection using IMU utilizes the redundant observations of GNSSs and IMUs on the same carrier motion to identify spoofing attacks. Since the IMU is relatively difficult for attackers to remotely tamper with, its measurements can serve as an anchor point to verify the authenticity of GNSS data. In most GNSS/IMU-integrated navigation systems, loosely coupled or tightly coupled filtering is usually adopted, and statistics such as innovations are used for spoofing detection. However, the focus here is on methods that do not rely on filter innovations but directly compare the output consistency of the two types of sensors. According to whether kinematic state estimation is performed, such methods can be further subdivided into two categories: direct comparison of raw observations and comparison based on kinematic constraint estimation.

4.1.1. Consistency Comparison of IMU Raw Data

This type of method directly uses statistical means to compare IMU raw data. GNSSs can derive acceleration through position solution, but carrier acceleration has random and unpredictable characteristics. Attackers find it difficult to forge acceleration sequences in real time that are highly consistent with true motion, so acceleration becomes a commonly used observation quantity. Its core idea is to obtain the GNSS acceleration sequence via second-order difference in GNSS positions, and perform a similarity check with the IMU-measured acceleration sequence to identify spoofing. Figure 6 illustrates the basic principle of spoofing detection based on acceleration sequence consistency.

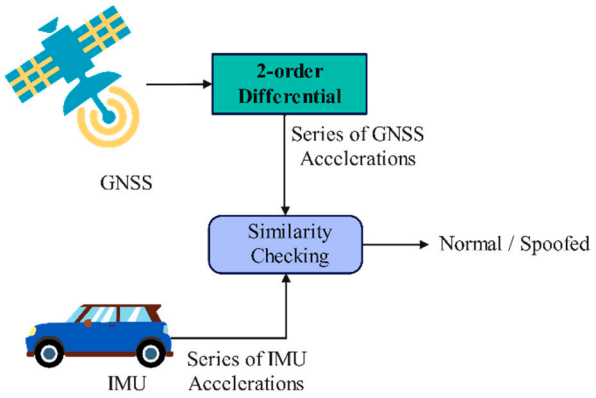


Figure 6. Spoofing interference detection based on acceleration sequence consistency.

Zhang Cheng et al. [55] filtered the effective frequency components of the acceleration sequence through frequency-domain analysis and verified the consistency of the acceleration sequences derived from IMUs and GNSSs by combining dynamic time warping

similarity and time delay analysis. When the sequence similarity is lower than the set threshold or the time delay exceeds a reasonable range, it is judged as spoofing interference. This type of method is simple to implement, has outstanding anti-forgery capability, and has no accumulated error, but it is highly dependent on the motion state, is susceptible to noise and disturbances, and has a high false alarm rate in complex environments.

4.1.2. Comparison Based on Kinematic Constraint Estimation

This type of method utilizes IMU data combined with kinematic equations or data fusion to generate reference values for the carrier state, and then performs consistency verification with the corresponding quantities solved by GNSSs. Its essence is to construct a hypothesis testing problem based on physical constraints. Ceccato et al. [56] proposed a Generalized Likelihood Ratio Test (GLRT) that uses raw observation data from IMUs and GNSSs, combines kinematic equations, and estimates the true trajectory through the maximum likelihood method, finally constructing a likelihood ratio statistic for spoofing judgment. The two-layer detection model proposed by Xie et al. [57] uses statistical features within a sliding window for rapid coarse screening in the first layer. If no trigger occurs, it proceeds to the second layer for position-attitude consistency check, comparing the deviations of latitude and longitude positions and heading angles (under high dynamics) solved by GNSSs and IMUs. When no spoofing is detected in both layers, the accumulated drift of the IMU is corrected using credible GNSS data through a Kalman filter, forming a detection-correction closed loop. The aided receiver integrity monitoring algorithm (ARIM) proposed by Shankar O T et al. [58] utilizes the state vector of INS to predict the current receiver position and velocity, calculates residuals with GNSS pseudorange and pseudorange rate observations, and eliminates satellite measurements exceeding the threshold, thereby achieving consistency verification at the pseudorange level. Compared with consistency comparison based on IMU raw data, this type of method has stronger physical constraints and is more robust to slowly varying spoofing attacks.

4.2. Spoofing Detection Methods Based on Communication Signal Consistency

Spoofing detection methods based on communication signal consistency focus on using communication interaction information to provide external verification references for GNSS positioning. Currently, such research, according to different application scenarios, shows platform-specific characteristics. Among them, Vehicle-to-Everything (V2X) and Unmanned Aerial Vehicle (UAV) Swarms, due to their rich communication link redundancy and collaborative interaction foundation, have become two core research scenarios; related detection methods are deeply explored on these two platforms:

4.2.1. Detection Methods Based on V2X

V2X, as shown in Figure 7, includes V2V (Vehicle-to-Vehicle), V2I (Vehicle-to-Infrastructure, such as RSU), V2N (Vehicle-to-Network), and V2P (Vehicle-to-Pedestrian). It is essentially a wireless communication system that enables vehicles to exchange information with all elements in the traffic environment, thereby achieving safer and more efficient driving, and is also a key technology supporting vehicle-road coordination and high-level autonomous driving. In terms of V2X-based spoofing interference detection, researchers provide external verification references for GNSS positioning by fusing collaborative information in V2X communication, effectively responding to the risk of a single source being hijacked.

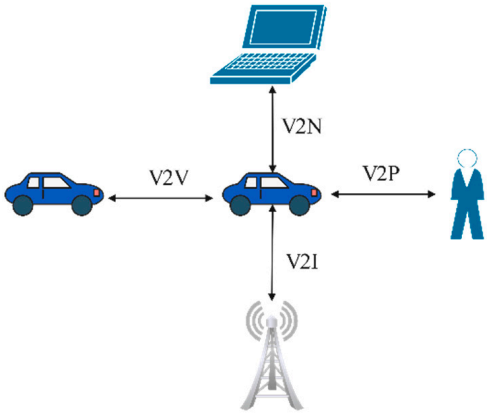


Figure 7. V2X.

Wenjie Liu et al. [59] proposed a probabilistic detection method based on opportunistic information (PDS), which fuses Wi-Fi, cellular network positioning, and vehicle inertial data. By using motion constraint regression and Gaussian processes to construct confidence intervals, and then using a likelihood ratio to detect spoofing attacks, this method outperforms single-source-based schemes on multiple datasets. Song Xiulan et al. [60] proposed a detection method based on adaptive Distributed Kalman Filter (DKF), which introduces online estimation of noise statistical characteristics, uses V2V communication of connected vehicles to obtain neighbor vehicle states, constructs adaptive DKF state residuals for chi-square detection, and can effectively identify covert false injection attacks and reduce false alarm rates. Gabriele Oligeri et al. [61] proposed a detection method based on crowdsourced information, which fuses broadcast signals from cellular base stations and Wi-Fi access points to estimate vehicle positions, and constructs abnormality judgment criteria by comparing with GPS positions. This method can achieve low false alarm rates and adjustable detection delays in various environments.

4.2.2. Cluster Verification Methods for UAV Swarms

The GPS spoofing detection method based on UAV swarms leverages the relative consistency among swarm members, namely the relative geometric positions and motion constraints between each UAV. Under normal conditions, UAVs in the swarm maintain a stable relative topological relationship. A spoofing attack will tamper with GNSS solutions, break the original geometric constraints of the swarm, and cause abnormal distortion of the relative position relationship, as shown in Figure 8. Through cross-validation among members and fusion of multi-source navigation data, the inconsistency between the GPS information reported by UAVs and their actual relative states can be identified, thus effectively detecting spoofing attacks.

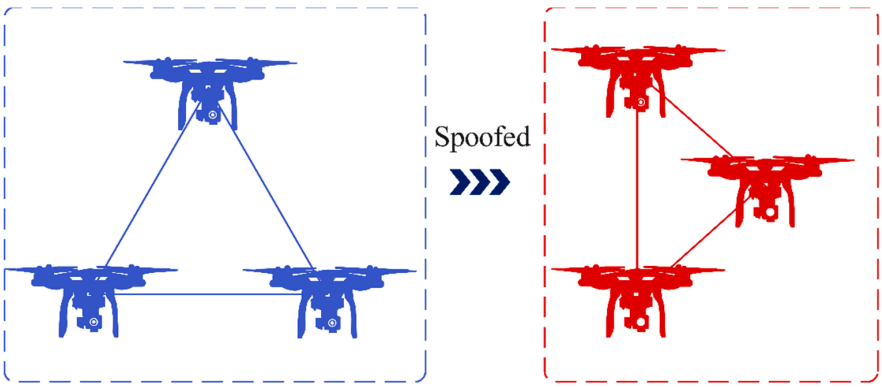


Figure 8. Changes in relative consistency of UAV swarms before and after spoofing.

Current detection methods based on UAV swarms mainly revolve around two core ideas: First, using the relative position relationships within the swarm as invariants for consistency verification, such as the mechanism proposed by Mykytyn et al. [62], which compares the distances between swarm members calculated based on GPS coordinates with the actual physical distances obtained by IR-UWB ranging. If the difference exceeds a threshold, spoofing is determined. The ASD method proposed by Lianxiao Meng et al. [63] targets small UAV swarms and uses swarm collaborative positioning and heading angle consistency to detect GPS spoofing. Its SSD mechanism designed for single-source spoofing utilizes the GPS position information broadcast by swarm members. When multiple members broadcast the same or abnormally overlapping GPS positions, an alarm is triggered. Second, deeply fusing multi-source sensor information with swarm collaborative deduction, such as the framework proposed by Fan et al. [64], which designs a position deduction driver (PID) for each sensor, uses on-board multi-sensors (IMU, RSSI, etc.) and collaborative data from neighboring UAVs, and generates high-confidence position estimates through ES EKF fusion to compare with GPS readings for spoofing identification. The RSOM mechanism proposed by Lianxiao Meng et al. [64] for multi-spoofing-source scenarios constructs a "virtual center aircraft" model based on magnetometer data from all members, compares the yaw angle deviation of each member with that of the virtual center aircraft, and marks the safest member as the swarm leader to restore navigation.

An in-depth analysis reveals a fundamental difference in the underlying mechanisms between the aforementioned UAV swarm cooperative detection and the "single-UAV communication-assisted" detection, which utilizes ground communication base stations (such as 5G). Single-UAV communication assistance essentially introduces another external absolute coordinate source independent of GNSSs (namely, infrastructure-to-vehicle or I2V communication) for cross-verification. In contrast, swarm cooperation completely eliminates the reliance on external absolute benchmarks. Its core lies in "endogenous relative state verification," which constructs a topological constraint network unrestricted by global geographic coordinate systems, relying solely on local relative information exchanged among members, such as mutual ranging and headings (namely, vehicle-to-vehicle or V2V communication). In recent years, research on communication-assisted GNSS spoofing detection for single UAVs has been dominated by artificial intelligence methods, with traditional detection methods being relatively limited. Among these, the method proposed by Famili et al. [65] is representative. This approach independently calculates the user's position using GPS and 5G Observed Time Difference of Arrival (OTDOA) technologies and computes the Euclidean distance between the two solutions. If this distance exceeds a predefined threshold of 20 m, GPS spoofing is detected, and the system can switch to 5G positioning to achieve attack evasion.

4.3. Spoofing Interference Detection Based on Visual Consistency

Spoofing interference detection technology based on vision mainly detects spoofing attacks by leveraging the physical consistency between camera video streams and satellite navigation information. The VISAS (Visual Integrity and Spoofing Attack Detection System) algorithm proposed by Davidovich et al. [66] first extracts SURF (Speeded Up Robust Features) features from consecutive video frames and calculates image similarity, then establishes a linear regression model between frame correlation and GPS displacement distance, and finally makes a judgment by comparing the deviation between the model-predicted distance and the actual GPS-measured distance. Its validation demonstrated the method's detection capability for 1–4 m position offsets in urban environments. Petro et al.'s [67] method focuses on heading consistency verification: it estimates the relative motion direction by obtaining the essential matrix from monocular images and propagates

the UAV attitude only through relative rotation between image frames to avoid relying on contaminated IMU data. Subsequently, it constructs HODs (Histogram of Orientations) for both GPS and image motion directions and calculates the Pearson correlation coefficient between them to detect spoofing. Gu et al. [68] proposed a detection algorithm based on data fusion, which uses an improved pyramid optical flow method to estimate the aircraft's horizontal velocity from video frames, fuses IMU data with a Kalman filter to calculate the optimal velocity estimate, and compares it with the velocity solved by GNSSs; meanwhile, it compares the altitude measured by IMU with the altitude solved by GNSSs. The abnormality of GNSS data is judged by the differences in these two dimensions: velocity and altitude.

In addition to IMU, communication signals, and vision, some researchers also combine other sensors to detect spoofing, such as altimeters and radars. Adolph et al. [69] utilize the consistency of barometric altitude, radar altitude, and terrain database to detect GPS spoofing: under normal conditions, barometric altitude equals the sum of radar altitude and terrain height. When spoofing occurs, incorrect GPS position leads to terrain height mismatch, and if the deviation filtered exceeds the threshold, it is judged as spoofing, which can detect offsets of more than 20 m. The distributed radar tracking and track fusion framework proposed by Pardhasaradhi et al. [70] detects inconsistencies by comparing the primary GPS data reported by UAVs with the secondary data tracked by radar, and uses track association algorithms to detect inconsistencies. After an attack is detected, it feeds back the radar fusion state to the UAV to restore navigation.

To systematically sort out the research progress of multi-source data consistency verification in GNSS spoofing detection, Table 3 categorizes and summarizes representative literature from dimensions such as comparison parameters, auxiliary sensors, application platforms, and decision methods.

Table 3. Summary of literature on GNSS spoofing detection methods based on multi-source data consistency verification.

Literature	Comparison Parameters	Platform	Detection Method
[55]	Acceleration	AV	Determine spoofing based on the similarity of acceleration sequences and time delay exceeding the threshold
[56]	Position, acceleration, initial velocity; displacement residual MAD, speed/acceleration variance	Vehicle	Construct a generalized likelihood ratio statistic and compare with the threshold
[57]	Ratio, position/speed mean slope, latitude/longitude position deviation, heading angle deviation	UAV	First layer: the sum of two combined indicators exceeds the threshold, respectively; second layer: latitude/longitude/heading angle deviation exceeds the threshold; two layers in series
[58]	Pseudorange residual, pseudorange rate (deltarange) residual	Launch vehicle	Eliminate the satellite measurement if the residual exceeds the threshold
[59]	GNSS observed position, mean and variance of multi-source fusion prediction	Vehicle	Calculate the log-likelihood value and compare with the Neyman-Pearson threshold
[60]	Adaptive DKF state residual (position, speed, acceleration)	Connected vehicle	Determine attack if the state residual chi-square statistic exceeds the threshold

Table 3. *Cont.*

Literature	Comparison Parameters	Platform	Detection Method
[61]	GNSS observed position, weighted centroid position of cellular base station/Wi-Fi access points	Connected vehicle	Joint configuration of single error threshold and continuous abnormal duration threshold, determine spoofing if timeout
[62]	Inter-aircraft distance calculated by GPS coordinates and measured by IR-UWB	UAV swarm	Determine spoofing if the distance difference exceeds the threshold (about 5.2 m) or GPS distance ≈ 0 and UWB distance > 0
[63]	SSD: difference between GPS broadcast position and geometrically inferred position; RSOM: difference between individual yaw angle and virtual center aircraft yaw angle	UAV swarm	SSD determines spoofing if the broadcast position is the same or the inferred position deviation exceeds the threshold; RSOM selects the member with the smallest yaw angle deviation as the safe aircraft, whose information is credible to assist the swarm recovery
[64]	IMU recursive position, RSSI ranging positioning, neighbor shared inferred position, GPS observed position	UAV	Obtain a credible position through multi-sensor + neighbor data fusion and compare with GPS; determine spoofing if the deviation exceeds the threshold or the confidence is too low
[65]	GPS and 5G positioning results	AV, UAV	If the Euclidean distance between GPS positioning and 5G positioning results is greater than the preset threshold, it is determined as GPS spoofing
[66]	Displacement distance predicted by video frame similarity, actual displacement distance measured by GPS	UAV	Determine spoofing if the deviation between predicted distance and actual distance exceeds the threshold
[67]	GPS motion direction and image motion direction	UAV	Determine spoofing if the Pearson correlation coefficient of the constructed HOD is lower than the threshold
[68]	Horizontal speed, ground clearance	UAV	Directly compare the absolute difference with the threshold (camera); if the residual, after first-order filtering and 10 s moving average, exceeds the 15 m threshold, it is judged as spoofing
[69]	Barometric altitude, radar altitude, terrain database query altitude	Aircraft	If the residual, after first-order filtering and 10 s moving average, exceeds the 15 m threshold, it is judged as spoofing
[70]	Position, speed, and covariance of GNSS output and radar observation	UAV	Track-track association (multi-dimensional S-D allocation + likelihood ratio test); if association fails, it is judged as spoofing

Overall, multi-source consistency check methods feature a lightweight architecture, with most schemes offering low computational complexity, high real-time performance, and ease of engineering deployment, serving as an important complement to filtering-based detection methods, particularly for mainstream application platforms such as vehicular platforms and UAVs; however, this class of technology has clear performance boundaries and common bottlenecks, with its detection reliability highly susceptible to three key factors. First is the time synchronization error between sensors; when sampling moments across

different sensors deviate and are not strictly aligned with timestamps, measurements during motion introduce systematic pseudo-biases, which directly disrupt the consistency check baseline and easily trigger false alarms. Second is the dynamic fluctuation in measurement uncertainty; the noise levels and measurement accuracies of individual sensors are not static; for instance, the increase in IMU random walk errors in highly dynamic scenarios or the intensified ranging fluctuations of communication signals in obstructed environments can cause the baseline of the consistency check to drift, thereby degrading the reliability of the detection results. Third is the interference from external environmental factors; vision-based methods are susceptible to changes in illumination, weather, and scene texture richness, whereas communication-based methods experience a significant decline in signal quality in urban canyons or densely built environments, both of which can lead to the failure of the consistency check logic and result in missed detections. Furthermore, most existing schemes still employ fixed threshold configurations and lack the capability to dynamically adjust decision criteria according to varying operating conditions and environments, causing false alarm and missed detection rates to further escalate in highly dynamic real-world scenarios, which marks a critical direction for future research.

Concurrently, driven by the continuous improvement of hardware computing power, the expansion of training datasets, and the rapid development of deep learning algorithms, artificial intelligence has gradually emerged as a mainstream methodology in the field of GNSS spoofing detection, owing to its powerful nonlinear modeling and adaptive learning capabilities.

5. AI-Based Spoofing Interference Detection

AI-based spoofing interference detection technology, leveraging its powerful nonlinear fitting capabilities and high-dimensional feature mining advantages, has become a key means to enhance the security of satellite navigation systems. Facing increasingly covert and complex spoofing attack patterns, current AI-based detection methods mainly focus on three strategies: First, the prediction-comparison-based strategy, which uses temporal models such as recurrent neural networks (RNN) to predict the carrier's motion state or signal features, and identifies anomalies by comparing the residuals between the model-predicted normal evolution trajectory and actual observed data. This method excels at capturing progressive offsets and consistency disruptions caused by spoofing attacks. Second, the classification-based strategy, which focuses on extracting highly discriminative feature vectors from multi-source heterogeneous data (such as sensor data, radio signal features, and control semantic features), and directly conducts end-to-end binary classification of the current state using a classifier. It has the characteristics of rapid response, strong feature interpretability, and sensitivity to single-point abrupt attacks. Third, the "unsupervised anomaly detection-based" strategy, which does not rely on pre-labeled spoofing samples, but learns the potential distribution of normal data through methods such as autoencoder reconstruction and clustering analysis, and judges isolated points or abnormal clusters that deviate from the normal distribution as attacks. It shows unique adaptability and robustness when dealing with new and unknown spoofing attacks.

5.1. Spoofing Interference Detection Based on Prediction Comparison

The core idea of GNSS spoofing interference detection methods based on prediction comparison is to use reliable information sources unaffected by spoofing attacks (such as inertial sensors, visual information, or physical models) to construct an "expected truth", and then compare it with possibly tampered GNSS observations or their derivatives (such as trajectories, speeds, residuals) for consistency, thereby identifying spoofing behavior.

Existing research literature shows diversified development trends in prediction targets, technical means, and comparison mechanisms.

In terms of motion trajectory prediction, most research uses vehicle inertial and vehicle sensors as reliable benchmarks to achieve displacement and trajectory prediction through deep learning models. Mohammadi et al. [71] fused speedometer and gyroscope data, used a Deep Neural Network (DNN) to predict inter-frame displacement and compared it with GNSS displacement, and effectively detected small cumulative offset attacks by leveraging the characteristics of the Gaussian distribution. Dasgupta et al. [72] used Long Short-Term Memory (LSTM) to fuse multiple motion parameters to predict displacement and combined Dynamic Time Warping (DTW) and k-Nearest Neighbors (k-NN) to detect steering behaviors, which can identify various attack types such as stop, overshoot, and wrong steering. Song et al. [73] addressed the low-cost IMU drift problem, used a hybrid model of Particle Swarm Optimization Temporal Convolutional Network (PSO-TCN) and Interacting Multiple Model Adaptive Kalman Filter (IMM-AKF) to suppress inertial navigation drift and generate virtual GNSS trajectories, and improved the robustness to slow drift attacks through fuzzy logic for spatiotemporal matching. Jiang et al. [74] proposed the DeepPOSE framework, which directly predicted speed and heading through Convolutional Neural Networks and recurrent networks, reconstructed the real trajectory and compared it with the GNSS trajectory, and eliminated cumulative errors by combining map alignment to achieve high-precision detection of trajectory-level spoofing. For low-injection-rate spoofing with extremely high concealment, fixed thresholds often fail to balance false alarm rate and missed alarm rate. Mohammadi et al. [75] innovatively introduced an unsupervised adaptive Density-Based Spatial Clustering of Applications with Noise (DBSCAN) clustering algorithm under the framework of using Multilayer Perceptron (MLP) to predict displacement, targeting subtle cumulative small bias attacks. This method does not require pre-labeled spoofing samples and only uses the recursive mean and standard deviation of non-anomalous data to dynamically adjust the detection threshold, greatly improving the interception probability of induced slow spoofing while maintaining a low false alarm rate. This type of method is intuitive and easy to understand, does not rely on filtering models, but is susceptible to sensor noise and cumulative errors.

In terms of communication collaborative prediction, prediction comparison methods relying on V2X and Integrated Sensing and Communication (ISAC) technology have been expanded. Ali Krayani et al. [76], based on the ISAC concept, used a coupled generalized dynamic Bayesian network to learn the coupling relationship between vehicle radio frequency signals and GNSS trajectories, and completed joint prediction and comparison of signals and trajectories by roadside units, which can simultaneously identify GNSS spoofing and V2I interference, suitable for V2X collaborative scenarios, but depends on roadside facility deployment.

In addition to trajectory prediction, prediction comparison methods based on filter innovation and residuals are more sensitive to slow spoofing. Zhong et al. [77] constructed compensated innovation and used Least Squares Support Vector Machine (LSSVM) for online prediction, improved the test statistic with predicted innovation, and enhanced the detection accuracy of single-channel spoofing in tight combinations. Zhu et al. [78] constructed a healthy residual generator for UAV platforms, learned the mapping relationship between reliable physical quantities and residuals through neural networks, and realized attack detection based on the deviation between predicted residuals and actual residuals. Zhang et al. [79] used a Backpropagation (BP) neural network to predict the statistical deviation caused by INS speed errors, corrected the observation data, and then conducted detection, effectively suppressing the interference of inertial navigation errors on detection performance. This type of method is highly sensitive to slow spoofing but

needs to rely on integrated navigation filtering models. To intuitively present the research status of predictive comparison-based spoofing detection methods, Table 4 summarizes the key information of relevant literature.

Table 4. Summary of literature on GNSS spoofing detection methods based on predictive comparison.

Literature	Model	Platform	Dataset	Prediction Accuracy	Detection Accuracy
[71]	MLP	AV	HDD	/	Recall: 95.23–100%; Accuracy: 97.83–100%
[72]	LSTM; k-NN&DTW	AV	HDD	RMSE = 0.02 m; Maximum absolute error = 0.5 m	Recall: 80–100%; Accuracy: 86–100%; F1: 0.9–1
[73]	PSO-TCN	Vehicle, AV	Open-source vehicle dataset	RMSE: East 0.88 m, North 2.10 m; Maximum deviation: East 1.83 m, North 3.67 m;	Detection rate: 100%
[74]	ConvLSTM	AV, UAV, ships, mobile platforms	BDD-100K + custom dataset	Velocity estimation error: as low as 5%. Displacement estimation error: average 26.8 m ($\pm$5.2 m, BDD-100K), 36 m (custom dataset)	Detection rate: 88%; False alarm rate: 4%
[75]	MLP + DBSCAN	UAV	HDD	/	Detection rate: 97.62–100%
[76]	C-GDBN	vehicle	NGSIM	RMSE (trajectory): 0.145–0.185 m; RMSE (RF signal): 1.89–1.99 m	Detection rate: 85–100%
[77]	LSSVM	Airborne	Simulated data	/	Detection rate: 90% (false alarm rate set to 0.01)
[78]	LSTM	UAV	Simulated data	/	/
[79]	BP	Airborne	Simulated data	Predicted innovation 57.66 m > filtered innovation 31.84 m (0.5 m/s spoofing rate, 300 s moment)	Detection rate 100% (Spoofing rate 0.4–0.9 m/s)

Combined with the content above and Table 4, it can be seen that existing predictive comparison-based spoofing detection methods are evolving from simple physical quantity prediction to complex trajectory reconstruction and prediction of the internal statistical characteristics of filters. In terms of prediction methods, they are developing from traditional analytical models and simple neural networks to deep learning models such as LSTM and Temporal Convolutional Network (TCN) to better handle nonlinear and temporal features; in terms of comparison objects, they are expanding from single displacement or velocity comparison to multi-dimensional comparisons such as residual norm and innovation sequence; in terms of response to attack strategies, the detection capability for covert spoofing attacks such as slow drift and small bias is continuously being enhanced.

5.2. Classification-Based Spoofing Interference Detection

5.2.1. Traditional Machine Learning

Traditional machine learning methods primarily rely on manually designed features and classical algorithms to map input data to classification labels, thereby effectively distinguishing between normal and spoofing states based on data-driven patterns. Such

methods do not depend on deep neural networks, featuring lightweight models and efficient inference.

Emre et al. [80] proposed an XGBoost-based GPS spoofing detection method for autonomous vehicles. Through data preprocessing and 10-fold cross-validation, they compared multiple models including decision trees, naive Bayes, and Multilayer Perceptron (MLP). It was found that XGBoost, with its ability to handle complex nonlinear data, achieved the highest accuracy of 97.64%. In the UAV scenario, the team led by Wei Xiaomin conducted a series of studies: MagDet [81] innovatively utilized the correlation between geomagnetic field anomalies and position, extracting statistical features of position and magnetic field strength. After comparing multiple algorithms such as Support Vector Machine (SVM), k-Nearest Neighbor (KNN), and Random Forest (RF), XGBoost was found to have the best detection performance, with a detection rate exceeding 99.5%. PerDet [82], based on the complementarity principle of sensor data, selected data from accelerometers, gyroscopes, magnetometers, GPS, and barometers as features. Using random forest for feature importance evaluation and selection, experiments showed that both Random Forest (RF) and XGBoost achieved an accuracy of 99.69%, outperforming the JSA method that only uses distance and angular velocity features. ConstDet [83] further introduced control semantics, selecting flight data from the horizontal position control process and altitude control process as features. After training with multiple machine learning algorithms and selecting the optimal classifier, experiments indicated that XGBoost was the best, effectively detecting GPS spoofing attacks with a detection rate of 97.70%. To intuitively present the research status of machine learning classification-based GNSS spoofing detection methods, Table 5 summarizes the key information of relevant literature.

Table 5. Summary of literature on GNSS spoofing detection methods based on machine learning classification.

Literature	Model	Platform	Dataset	Detection Accuracy
[80]	XGBoost	AV	AV-GPS-Dataset-1	Average accuracy: 97.644%; Accuracy standard deviation: 6.625%
[81]	XGBoost	UAV	Self-collected dataset	Accuracy: 88.50–99.57%; F1 value: 85.97–99.55; Equal Error Rate (EER): 0.49–0.51%; Accuracy: 99.69%, Precision: 99.07%/98.76%; Recall: 99.38%/99.69%, F1 value: 9.22%
[82]	RF, XGBoost	UAV	Self-collected dataset	Accuracy: 99.69%, Precision: 99.07%/98.76%; Recall: 99.38%/99.69%, F1 value: 9.22%
[83]	XGBoost	UAV	Self-collected dataset	Accuracy: 97.70%; Recall: 96.76%

5.2.2. Deep Learning

Unlike traditional machine learning methods that rely on manually designed features and shallow models, deep learning methods achieve autonomous extraction of high-order abstract features from raw data through deep neural networks, enabling efficient learning of complex nonlinear mapping relationships and better adapting to the demands of GNSS spoofing detection in complex dynamic environments.

Navigation parameters generally exhibit temporal features, and numerous studies leverage single temporal neural networks to mine the temporal features of navigation data for high-precision spoofing identification. Dang et al. [84] compared the similarity between the GPS position topology and communication topology of UAV swarms using Graph Neural Networks (GNN), while simultaneously utilizing Recurrent Neural Networks (RNN) to analyze the temporal variation in this similarity. If the similarity remains below a threshold continuously, it is determined that the UAV swarm is under GPS spoofing

interference. For single-machine slowly varying spoofing, Guizzaro et al. [85] used an IMU and a GNSS receiver to estimate velocity, and calculated the normalized Euclidean distance of the velocity difference within a fixed time window. The sequence of normalized Euclidean distances within a window was taken as the feature vector, which was then input into a three-layer feedforward neural network for binary classification to output whether a spoofing attack was occurring. Raj et al. [86] utilized Bidirectional Long Short-Term Memory (Bi-LSTM) networks to detect GPS spoofing points in AIS data, inputting the historical trajectory sequence of ships into Bi-LSTM. The model learns normal navigation patterns from both forward and backward directions, outputting an anomaly score for each AIS message; if the anomaly score exceeds a preset threshold, the point is determined to be a spoofed false position. Wang et al. [87] addressed the challenge of attack detection for quadrotor UAVs under nonlinear dynamics and non-Gaussian noise, proposing the QUADFormer framework. This framework first generates residual sequences sensitive to anomalies using an Extended Kalman Filter (EKF), then captures temporal features through a Transformer architecture for classification detection, and triggers a resilient state estimation mechanism upon detecting an attack. Experiments prove its detection performance is superior to methods such as CUSUM, SVM, CNN, and LSTM. Jiang et al. [88] preprocessed the real-time dashcam image sequence of a vehicle, and the Google Street View (GSV) reference image sequence queried from the reported GPS coordinates, inputting them into a Gated Recurrent Unit (GRU) to extract temporal features, and then used a binary classifier to determine whether the two sequences match. If they do not match, it is determined to be a GPS spoofing attack.

In addition to mining navigation temporal features, the detection approach relying on cellular network signal features has also been widely applied to UAV GPS spoofing detection. The core principle is to identify abnormal spoofing signals by utilizing the difference feature between the theoretical path loss from signal theory and the actual measured path loss. Dang et al. systematically studied the problem of cellular network-assisted UAV GPS spoofing detection. Specifically: (1) Directly extracting statistical features of the path loss difference and using these statistical features as input to train an MLP model, which outputs whether the current GPS position is spoofed [89]; (2) Constructing a deep ensemble learning framework, where each base station independently processes statistical features with an MLP to obtain local predictions, and then integrates multi-base station predictions through threshold-weighted aggregation or meta-learners [90]; Introducing a 3D radio map to provide a theoretical signal strength (RSS) baseline, using the difference between theoretical and measured signal strength as features for classification, and also utilizing particle filtering to relocalize the UAV, with errors controllable within 10 m [91].

Although artificial intelligence-based detection models perform excellently on specific datasets, existing models often face generalization failures when applied across different scenarios. The core reasons for this bottleneck mainly center on three aspects: first, the high dynamics of real physical environments lead to dynamic domain shifts, which cause deviations in feature distributions between training data and actual operational data; second, the extreme scarcity of high-quality and diverse spoofing attack samples makes it difficult for models to learn complete boundaries of attack features; and third, issues such as domain mismatches of sensor noise exist across different carrier platforms and heterogeneous hardware. To further enhance detection accuracy and model generalization ability in complex scenarios, federated learning and transfer learning have become important evolutionary paths. Chai et al. [92], based on a federated learning framework, used a TCN-Transformer hybrid neural network to perform anomaly detection on UAV navigation temporal data. Among them, the Temporal Convolutional Network (TCN) extracts local temporal features through causal convolution and dilated convolution, while

the Transformer captures global dependencies using a self-attention mechanism. Finally, a classifier outputs whether the current signal is normal, spoofed, or interfered. Multiple ground station clients train models locally and only upload parameters to the central server for accuracy-weighted aggregation, achieving high-precision detection while protecting data privacy. In terms of cross-domain generalization, Dang et al. [93] utilized transfer learning to transfer the parameters of a pre-trained Convolutional Neural Network (CNN) model between different edge servers. When a UAV switches base stations, retraining is not required. The CNN takes the sequence of differences between the theoretical and measured path loss between the base station and the UAV as input, automatically extracts deep features, and classifies to determine whether the GPS position is spoofed.

Furthermore, to adapt to the lightweight and real-time detection requirements of embedded and mobile devices, many studies focus on constructing lightweight detection models. Alsarhan et al. [94] calculated the drift index of dual GPS (GDI), combined with features such as quaternions and GPS accuracy, and, after Kalman filtering and normalization preprocessing, input the data into a lightweight neural network automatically searched by DARTS (Differentiable Architecture Search) for classification. This method enables real-time spoofing detection. The GPS-IDS framework proposed by Abrar et al. [95] has significant advantages in model architecture lightweighting. Unlike traditional methods that rely on long-term sequence caching and complex network structures (such as LSTM), this framework only inputs 14-dimensional features at a single moment, without the need for historical sequence accumulation, yet it can characterize the transient physical behavior of the vehicle. Combined with lightweight classifiers such as Multilayer Perceptron (MLP), the framework significantly reduces computational complexity while maintaining a high detection rate, with a detection speed 56.5% higher than traditional EKF methods. To intuitively present the research status of GNSS spoofing detection methods based on deep learning classification, Table 6 summarizes the key information of relevant literature.

Table 6. Summary of literature on GNSS spoofing detection methods based on deep learning classification.

Literature	Model	Platform	Dataset	Detection Accuracy
[84]	GCN + LSTM/GRU	UAV	Simulated data	Detection rate > 90%
[85]	MLP	UAV, AV	Public database	When the minimum segment distance of the trajectory is equal to 10 m, both false alarm rate and miss detection rate can reach approximately 1%
[86]	Bi-LSTM	Ship	Marine Cadastre	Detection rate: 91%; Precision: 94%; Recall: 92%; F1: 93%
[87]	Transformer	UAV	Self-collected dataset	Precision: 89–92%; Recall: 93%
[88]	CNN/Transformer + GRU	Vehicle, AV	BDDG4k	Precision: 89–92%; Recall: 93–97%; F1: 91–94%; Accuracy: 82.5–94%; Precision: 78–92.3%; Recall: 90.6–96%; F1: 83.8–94.1%
[89]	WD-MLP	UAV	Simulated dataset	Single base station accuracy: 80%; Three base stations accuracy: 93%
[90]	MLP	UAV	Simulated dataset	Dual base station accuracy > 97%, single base station accuracy $\geq$ 83%; Miss detection rate (FN): <2%
[91]	MLP	UAV	Simulated dataset	/

Table 6. *Cont.*

Literature	Model	Platform	Dataset	Detection Accuracy
[92]	TCN-Transformer	UAV	Public data	Accuracy: 91.97%; Recall: 92.00%; Precision: 98.00%; F1: 91.00%
[93]	CNN	UAV	Simulated dataset	Accuracy: >88%; Precision: 93%; Recall: 79%; F1: 85%
[94]	DARTS	UAV	Public dataset	Accuracy: 99.99%; F1: 99.9%; AUC value: 99.99%
[95]	MLP	AV	AV-GPS-Dataset	Real data: Average F1 = 94.4%, FP R = 0.21%, FNR = 0.12%; Simulated data: Average F1 = 97.1%, FP R = 0.76%, FNR = 0%

5.3. GPS Spoofing and Jamming Detection Based on Unsupervised Anomaly Detection

Unsupervised anomaly detection-based GPS spoofing detection methods have a key characteristic: they only use sensor data from normal (non-spoofing) states for model training, without relying on any labeled attack samples. This paradigm is particularly suitable for the security field, as spoofing attack data is scarce, variable, and difficult to obtain in advance, while unsupervised learning can automatically extract features from normal behavior patterns and identify unknown attacks by detecting anomalies that significantly deviate from the learned patterns. The key challenge of this method lies in how to precisely characterize the boundary of normal behavior to reduce false alarms while maintaining sensitivity to subtle, gradual attacks.

Current related research has formed diverse technical solutions around unsupervised modeling, creating differentiated detection advantages for different scenarios and attack characteristics. Wang et al. [96] proposed an infrastructure-assisted detection framework, using roadside units (RSUs) as independent and trustworthy security data sources, relying on the Isolation Forest algorithm to identify inconsistencies between GPS measurement data and RSU measurement data. By introducing benchmark reference information that attackers cannot manipulate, the reliability of spoofing detection is significantly improved. Yang et al. [97] constructed an attention autoencoder model, focusing on the temporal correlation features of multi-sensor data, using an autoencoder to reconstruct normal temporal data and taking reconstruction error as the core basis for anomaly judgment; meanwhile, the attention mechanism adaptively focuses on feature dimensions sensitive to spoofing attacks, effectively eliminating interference from irrelevant features and improving detection accuracy in multi-source data scenarios. For the lightweight detection requirements of consumer-grade UAVs, Wang et al. [98] proposed the TimesNet model, innovatively converting GPS temporal signals to the frequency domain for analysis, extracting key frequency components and recombining multi-scale temporal features to adapt to complex flight dynamic scenarios, achieving lightweight real-time detection of GPS spoofing attacks.

In addition, many studies have achieved technical upgrades from dimensions such as privacy security, covert attack detection, and algorithm generalizability. Liu et al. [99] combined the advantages of unsupervised learning and federated learning to build a federated self-supervised detection framework, where each mobile platform generates pseudo-labels through traditional detection methods for local training of LSTM models and only uploads model parameters to achieve global collaborative optimization, effectively improving the detection performance of the global model while strictly protecting data privacy. For highly covert sub-perceptual slow drift attacks, Otieno et al. [100] designed a stacked LSTM variational autoencoder and its memory-enhanced variants, mining deep

latent representations of normal trajectories through deep temporal modeling, which can accurately capture long-term accumulated small trajectory deviations and address the shortcomings of traditional methods in identifying gradual covert spoofing attacks. The PADS framework proposed by Liu and Papadimitratos [101] integrates the dual advantages of model-driven and data-driven approaches, achieving position data smoothing through motion-constrained regression, completing data uncertainty modeling with Gaussian processes, constructing detection statistics by fusing multi-source confidence intervals, and finally making decisions based on a lightweight Loda detector, providing a lightweight unsupervised GPS spoofing detection solution for consumer-grade devices that requires no hyperparameter tuning and can output probabilistic results. To intuitively present the research status of GNSS spoofing detection methods based on unsupervised anomaly detection, Table 7 summarizes the key information of relevant literature.

Table 7. Summary of literature on GNSS spoofing detection methods based on unsupervised anomaly detection.

Literature	Model	Platform	Dataset	Detection Accuracy
[96]	Isolation Forest	Ground vehicles	Self-collected data, Simulated data	F1: 0.73–0.93; Recall: 0.67–1; Precision: 0.76–0.88
[97]	Autoencoder + Multi-head Attention Mechanism	AV	Self-collected data	Detection rate: 100%; False alarm rate: 0.57%
[98]	TimesNet	UAV	Self-collected data	Accuracy = 99.57%; Accuracy = 98.63%; Recall = 99.97%; F1 = 99.30%
[99]	LSTM-based self-supervised framework	Mobile phones, vehicles, UAV	Self-collected data	AUC: 86.7–87.4%
[100]	Stacked-LSTM-VAE, Mem-LSTM-VAE	AV	AV-GPS-Dataset1-Normal-Data	AUC: 85.8–99.3%
[101]	Loda	Mobile phones, AV	Apollo autonomous driving dataset, Simulated data, Self-collected data	True positive rate (urban environment): 98–99% (False positive rate 5–15%); True positive rate in the same false positive rate interval (rural environment) approximately 54–80%

Overall, AI-based detection methods have strong resistance to complex attacks and rich functional extensibility. However, prominent common problems exist: most models only achieve excellent metrics in simulated environments or specific self-built datasets. The standards for datasets, attack types, and noise conditions in different studies are not unified, and detection metrics lack cross-comparability, making it difficult to verify the generalization ability of models. At the same time, some complex deep learning models have large computational overhead and are not friendly for deployment on embedded terminals. Moreover, many methods lack stability when switching scenarios or changing environments. Subsequent research needs to focus on improving the cross-scenario generalization ability of models, unifying test standards, and promoting model lightweighting and engineering deployment optimization.

6. Research Status and Development Trends

Previous sections systematically reviewed the research status of multi-source fusion GNSS spoofing and interference detection from three technical paths: filtering algorithms, multi-source data consistency verification, and artificial intelligence. To further reveal the intrinsic characteristics of different methods and support subsequent challenge analysis and

future outlook, Table 8 summarizes and compares the representative subclasses, advantages, and limitations of the aforementioned various methods.

Table 8. Summary of advantages and limitations of multi-source fusion GNSS spoofing interference detection methods.

Detection Method	Subclass	Advantages	Limitations
Filtering Algorithms	LC	1. Simple structure 2. Low computational complexity	1. Poor detection robustness in complex spoofing scenarios 2. Low sensitivity to slow-varying spoofing detection
	TC	1. Directly utilize pseudorange/ pseudorange rate for earlier signal layer anomaly detection. 2. Combined with robust estimation improvements, effectively counteract filter tracking effects and shorten detection delay; 3. Can achieve channel-level spoofing satellite identification and isolation.	1. Depend on accurate system/measurement noise statistical models. 2. Improved algorithms increase computational complexity.
Consistency Verification	IMU	1. Strong anti-forgery capability 2. High autonomy	1. Strong dependence on motion state 2. Susceptible to sensor noise and vibration interference, high false alarm rate in complex environments
	Communication Signals	1. Equipped with communication functions, can use clusters for collaborative cross-validation; 2. The carrier only needs a communication module, low cost	1. Depend on communication infrastructure deployment (e.g., RSU) or cluster scale 2. Communication links may be interfered with or spoofed
	Vision	1. Low-cost cameras, widely mounted on UAVs/autonomous driving platforms; 2. Not affected by GNSS signal environment interference	1. Greatly affected by environmental factors such as lighting, weather, and texture
Artificial Intelligence	Prediction Comparison	1. No need for precise physical models; 2. Can control and mine temporal features of navigation data; 3. Can provide short-term navigation under GNSS denial	1. Depend on high-quality normal data training; 2. Prediction errors accumulate over time; 3. Complex model structure, difficult for embedded deployment; 4. Thresholds mostly inherit from traditional statistical methods
	Classification	1. End-to-end decision-making, fast response; 2. Flexible feature engineering, can fuse heterogeneous multi-source information	1. Heavily dependent on labeled spoofing samples, difficult to obtain; 2. Weak cross-scenario generalization ability, high risk of overfitting; 3. Poor interpretability
	Unsupervised Anomaly Detection	1. No need for labeled spoofing samples; 2. Good adaptability to unknown types of spoofing attacks	1. Depend on high-quality normal data training; 2. Strong scene migration ability; 3. Poor interpretability

6.1. Research Status

Based on the literature review and comparative analysis above, the field of multi-source fusion GNSS spoofing and interference detection currently presents the following five prominent research status aspects:

1. Application platforms are highly concentrated on UAVs and autonomous vehicles (AVs), while research on other carriers is relatively weak.

Most existing studies use UAVs and AVs as validation platforms, which is closely related to the high sensitivity of these two platforms to navigation security, rich sensor configurations, and extensive application scenarios. In contrast, research on multi-source spoofing detection for carriers such as aircraft, ships, and missiles is significantly insufficient.

2. Detection targets are increasingly focused on slow-varying spoofing interference, and countering covert attacks has become a core research focus.

Early studies mainly focused on abrupt or high-power spoofing attacks, which are easy to detect. However, slow-varying spoofing with more practical value induces receiver positioning results through long-term, small-step increments, effectively evading detection based on instantaneous statistical features. In recent years, a large number of works have been devoted to improving the sensitivity to slow-varying spoofing, such as using multi-epochs to detect spoofing in filtering-based methods and leveraging temporal and multi-epoch features to enhance sensitivity in AI-based methods. This reflects that the current research focus is shifting from "whether spoofed" to "whether able to identify slowly infiltrating spoofing".

3. AI-based methods have become the mainstream technical route, but engineering validation is still insufficient.

In terms of literature volume and research activity, AI-based spoofing detection methods now account for nearly half of the field. AI methods, with their strong nonlinear fitting and high-dimensional feature mining capabilities, show significant advantages in detection accuracy in complex scenarios and identification of covert attacks. However, most AI research relies on simulated data or self-built small-scale datasets, lacking unified public test benchmarks; the generalization ability of models across platforms and scenarios has not been fully verified; and the computational overhead and interpretability issues of deep models limit their deployment in real-time embedded systems. In contrast, although filtering-based and consistency verification methods have relatively declined in research popularity, their solid theoretical foundations and strong interpretability still make them irreplaceable in practical engineering.

4. Detection granularity is evolving from coarse-grained system-level alarms to refined perception and isolation.

Early multi-source spoofing detection often only answered the binary question of "whether spoofing exists". Current research is committed to providing more refined information: not only detecting attack occurrences but also determining the start and end times of attacks; not only reporting system anomalies but also locating specific spoofed satellite channels; not only completing detection but also recovering the true position after detection. This evolution from "existence or not" to "when, where, and how to recover" marks that spoofing detection is upgrading from a simple monitoring tool to a core component of system resilience construction.

5. Detection threshold setting mostly relies on experience or specific model assumptions, lacking adaptive mechanisms.

Most studies set detection thresholds based on experience or specific model assumptions. This static threshold strategy is difficult to maintain optimal detection performance when facing dynamically changing motion patterns, sensor noise drift, or different spoofing types. A few works have tried to introduce adaptive adjustment mechanisms, but overall, adaptive threshold design remains a relatively weak link in current research, restricting the robustness and transferability of methods in real complex environments.

6.2. Development Trends

1. Expand from "limited multi-source" to "elastic heterogeneous fusion" to break through the inherent limitations of single sensors.

Current multi-source fusion detection schemes are mostly limited to the binary architecture of GNSS + INS, or at most introduce two to three sensors such as vision and communication. This "limited multi-source" still has blind spots in specific scenarios: for example, long-term drift of INS, degradation of vision due to lighting, and communication signals relying on infrastructure. In the future, sensor types and fusion depth should be further expanded, incorporating more heterogeneous information sources such as LiDAR, geomagnetic fields, barometers, radar, and millimeter-wave into a unified detection framework. By constructing an "elastic heterogeneous fusion" system, credible sensor subsets can be adaptively selected according to environmental dynamics to achieve deep cross-validation of multi-source information, thus fundamentally improving immunity to complex collaborative spoofing from a physical level.

2. Constructing public, standardized, and high-quality datasets to support fair evaluation and benchmarking of algorithms.

In research on artificial intelligence-based multi-source fusion and spoofing detection, data is the core driver for model learning and algorithm validation. At this stage, comprehensive datasets specifically dedicated to navigation security are relatively scarce. Consequently, most studies rely on existing public datasets from fields such as autonomous driving, traffic flow observation, and maritime transport to perform theoretical validation. The typical open-source datasets frequently cited in this field, along with their key application characteristics, are summarized in Table 9.

Table 9. Comparison of typical open-source multi-source sensor datasets.

Dataset Name	Primary Target Platforms	Data Modalities and Key Features	Real Attack Labels?
HDD	Vehicles	Video streams, rich low-level CAN bus data (vehicle speed, yaw rate, steering wheel angle, acceleration, etc.)	No
BDD	Vehicles	Massive visual data (covering all-weather/multi-scenario), high-precision annotations (lane markings/drivable areas), basic GPS/IMU	No
NGSIM	Land traffic/Traffic flow	Bird's-eye view camera footage, 10 Hz high-precision microscopic trajectories (coordinates, velocity), multi-vehicle interaction behaviors (car-following, lane-changing)	Yes
AV-GPS-Dataset	Autonomous driving, UAVs	Multi-modal sensor alignment, low-level raw GNSS observations (NMEA, pseudorange, Doppler shift, SNR)	No
Marine Cadastre	Vessels	Massive historical AIS trajectory data, highly inertial navigation status, marine meteorology, and waterway geographic constraints	No
Apollo (ApolloScape)	Vehicles	3D LiDAR point clouds, High-Definition Map (HD Map), centimeter-level RTK-GNSS, tactical-grade IMU	No

As illustrated in Table 9, although existing public datasets provide a massive volume of heterogeneous sensor observations and authentic vehicle kinematic trajectories, the vast majority are general-purpose datasets that do not directly contain real labeled annotations of GNSS spoofing attacks. This status quo forces existing studies to often utilize these datasets only as normal physical baselines or rely heavily on post hoc manual injection of simulated attacks. Therefore, constructing public, standardized, and high-quality datasets is highly necessary to support the fair evaluation and benchmarking of different algorithms.

Currently, there is a severe lack of unified benchmark datasets in this field. The vast majority of studies rely on self-collected or simulated data with different spoofing types, vehicle dynamics, and environmental conditions, making it difficult to conduct fair and quantitative performance comparisons among different methods. In the future, there is an urgent need to construct public datasets that encompass a variety of spoofing attack types (including generative, repeater, progressive traction, and multi-satellite cooperative spoofing), multiple platforms (such as UAVs, vehicles, vessels, and aircraft), and diverse environments (such as urban, suburban, maritime, and high-altitude areas). These datasets should simultaneously contain clean, normal data and fully annotated spoofing data, while unifying evaluation metrics (such as detection rate, false alarm rate, detection latency, and computational overhead) to provide a solid foundation for algorithm validation and benchmarking.

3. Break through the generalization bottleneck of AI-based methods and develop advanced paradigms such as federated learning and transfer learning.

Although AI-based methods dominate current research, their insufficient generalization ability across scenarios and platforms remains a core obstacle to engineering deployment. A high-precision model trained on a specific dataset often experiences a sharp performance drop when the carrier, sensor, or environment changes. In the future, federated learning frameworks should be vigorously developed to achieve multi-client collaborative training while protecting data privacy and enhancing the global generalization ability of models; at the same time, transfer learning and domain adaptation technologies should be promoted to enable models to quickly adapt to new scenarios with a small number of target domain labeled samples or no labels. In addition, zero-shot learning and meta-learning also provide new theoretical paths for dealing with unknown spoofing attacks.

4. Move from theoretical validation to engineering deployment, focusing on lightweight and real-time requirements.

Most current multi-source spoofing detection research stays at the simulation validation or offline data processing stage, with a large gap from actual embedded real-time systems such as UAV flight controllers and vehicle ECUs. Future research should pay more attention to lightweight model design, reducing the computational and storage overhead of deep learning models through means such as model pruning, quantization, and knowledge distillation; at the same time, optimize the computational flow graph of detection algorithms and design real-time detection architectures with single forward inference or low-delay recursive updates. In addition, the tight coupling integration of detection modules with integrated navigation systems should be promoted, making spoofing detection an inherent part of the navigation solution rather than an external module, thus improving overall system resilience while ensuring real-time performance.

5. Focusing on differentiated application platforms and constructing customized detection pathways for specific scenarios.

Current academic research is highly concentrated on UAVs and autonomous vehicles, whereas research targeting the high-reliability requirements of aviation and maritime

navigation, as well as high-real-time scenarios like missiles, lags severely behind. Due to stark differences in sensor configurations, dynamic characteristics, and environmental constraints, differentiated technical tackling pathways must be implemented in the future:

(a) Autonomous and Intelligent Connected Vehicles (High-density interaction scenarios): Efforts should be directed toward the deep integration of 5G-A (5.5G) high-precision Integrated Sensing and Communication (ISAC) signals with Vehicle-to-Everything (V2X) networks. Roadside Units (RSUs) should be utilized to provide absolutely trustworthy spatiotemporal references, upgrading single-vehicle detection to "vehicle-road-cloud" crowd-intelligence cooperative cross-verification to resolve multi-path confusion and spoofing detection challenges in urban canyons.

(b) Swarm UAVs (Highly collaborative and flexible scenarios): This direction should completely eliminate the reliance on external absolute coordinate systems, focusing instead on developing relative spatial topology consistency checks based on inter-UAV data links such as UWB or ad hoc networks. By leveraging relative positions, heading angles, and motion constraints within the swarm as invariants, a distributed collective anti-spoofing immunity network can be constructed.

(c) Civil Aircraft and Missiles (Highly dynamic and high-real-time scenarios): For supersonic or highly maneuvering platforms, detection mechanisms based on vision or low-frequency environmental perception are highly prone to failure. Research should focus on breaking through ultra-fast closed-loop verification technologies based on high-frequency, tactical-grade INS. This should be combined with next-generation satellite navigation authentication regimes, such as BeiDou-3/BeiDou-4, and the assistance of strong signals from Low Earth Orbit (LEO) constellations, to achieve a hard cutoff within milliseconds of the spoofing signal pulling the tracking loop.

(d) Ocean-going Vessels (Long-endurance and low-dynamics scenarios): Vessels possess enormous inertia and slow motion states, making them highly vulnerable to undetected small-step ramp-type spoofing attacks. Conventional high-frequency dynamic verification should be abandoned in favor of utilizing marine radar, Doppler Velocity Logs (DVL), and independent AIS signals, or even integrating celestial navigation. This will allow the construction of long-endurance, low-frequency error suppression models and macro-waterway topology anomaly identification mechanisms.

7. Summary

This paper systematically reviews 67 representative studies in the field of multi-source fusion GNSS spoofing and interference detection from 2021 to 2026. It classifies existing methods into three major categories: filtering-based algorithms, multi-source data consistency verification-based methods, and artificial intelligence-based methods. It systematically compares different detection methods, summarizes the research status and development trends, and provides references for related research. Current research shows the status of platforms concentrated on UAVs and vehicles, detection targets focused on slow-varying spoofing, AI-based methods becoming the mainstream, detection granularity becoming refined, and threshold setting relying on models or experience. In the future, efforts should be made to focus on developing elastic heterogeneous fusion frameworks, building public and standardized datasets, breaking through the generalization bottleneck of AI-based methods, promoting lightweight and real-time deployment, and strengthening the research on spoofing detection for key platforms such as aircraft and ships.

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